Where Typhoon Exposure and Rice-Price Volatility Overlap

Stakeholder policy brief | Philippines | 2024-2025

THE QUESTION

Which Philippine regions and provinces had both more severe typhoon warnings and larger rice-price increases during 2024-2025?

THE MAIN MESSAGE

Some places stand out on both signals and should receive closer monitoring. This analysis identifies where the patterns overlap; **it does not say that typhoons caused rice-price changes.**

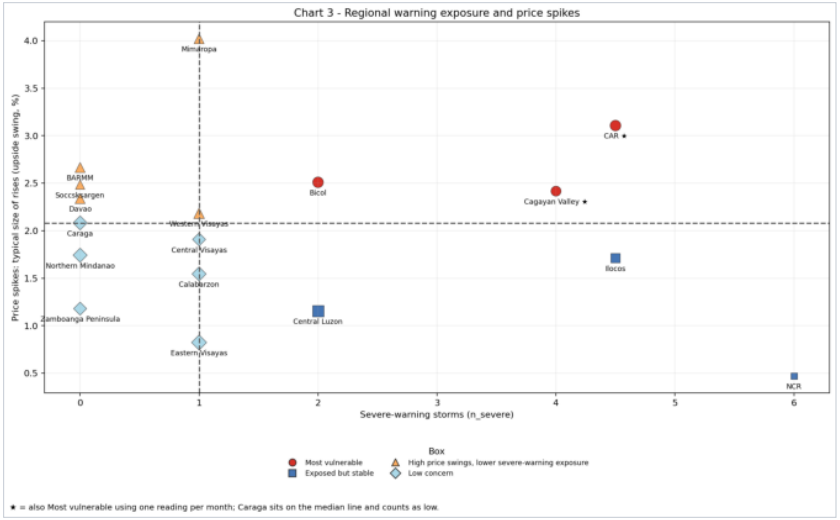
WHAT THE ANALYSIS FOUND

3 regions	Bicol, CAR, and Cagayan Valley were the regional "Most vulnerable" group in this descriptive screen.
12 provinces	were in the high-exposure / high-price-swing group (above the median on both KPIs).
67 of 81	provinces kept the same group in the one-reading-per-month robustness check (regions: 13 of 17).
75 of 81	provinces had rice prices end lower, but upward movements varied substantially.

HOW TO USE THIS BRIEF

Use the results to prioritize monitoring and decide where to collect more information. Do not use the group labels as proof of damage, household impact, vulnerability, or cause and effect.

REGIONAL COMPARISON



Bicol, CAR, and Cagayan Valley are above both regional medians.

HOW THE FOUR GROUPS ARE SET

The four groups are **descriptive screening labels**. Each region or province is placed using two KPIs: above or below the median **n_severe** (warning exposure) and above or below the median **upside_swing** (rice-price spikes). The annual severe-warning exposure rate is tracked separately and **does not replace** this two-axis segmentation.

	Below median n_severe	Above median n_severe
Above median upside_swing	△ High price swings, lower severe-warning exposure	● "Most vulnerable" (high exposure, high price swing)
Below median upside_swing	◇ Low concern	■ Exposed but stable

PROVINCES FLAGGED BY THE SCREEN (12)

Abra · Apayao · Aurora · Cagayan · Camarines Norte · Camarines Sur · Catanduanes · Ifugao · Ilocos Sur · Isabela · Mountain Province · Quirino

RECOMMENDED NEXT STEPS

1. **Monitor** the flagged regions and provinces, especially during and after severe-weather periods.
2. **Track over time** warning exposure (n_severe, annual rate) and rice-price movements (upside_swing), using the same KPIs as longer periods of data become available.
3. **Investigate separately** transport disruption, supply, demand, policy changes, and physical damage, including household food-security indicators.
4. **Avoid causal claims** the analysis cannot support; use the labels to decide where to collect more information.

WHAT THIS ANALYSIS CANNOT ESTABLISH

- × A typhoon caused a particular price increase.
- × A warning means a province experienced physical damage, household impact, or vulnerability.
- × A flagged province was harmed more than another province.
- × The observed overlap excludes other explanations.

THE THREE KPIs BEHIND THIS BRIEF

Each KPI is a monitoring measure of warning exposure or price movement, not of damage.

1 Severe-warning exposure

`n_severe`

What it measures. The number of distinct storms in which a province reached Wind Signal 3 or higher. It measures warning exposure, not physical damage.

How it is calculated. Count the distinct storms where the province reached Wind Signal 3 or higher. A storm is counted once per province.

Why it matters. It answers the "more severe typhoon warnings" half of the question and sets the median cut-off for the exposure axis.

In the data. 2,717 warning records processed; 63 provinces in at least one warning signal; 29 of 41 storms with provincial warning listings; 28 repeat severe-warning listings across 16 provinces.

2 Rice-price upside swing

`upside_swing`

What it measures. The typical size of upward percentage movements in regular-milled rice prices, using the 2024-2025 price readings.

How it is calculated. Take the percentage rise between consecutive readings, square each, average them, then take the square root (root-mean-square). Decreases and unchanged readings count as zero.

Why it matters. It answers the "larger rice-price increases" half of the question. Price decreases never raise the score, so only upward spikes count.

In the data. 75 of 81 provinces had rice prices end lower, but upward movements varied substantially. Five highest province scores: Masbate, Batangas, Marinduque, Occidental Mindoro, Aurora.

3 Annual severe-warning exposure rate

`severe_warning_rate`

What it measures. The percentage of distinct storms in a calendar year in which a province reached Wind Signal 3 or higher. It is a monitoring KPI, not a damage rate.

How it is calculated. Distinct storms in the year where the province reached Wind Signal 3 or higher, divided by the distinct storms in that year, times 100.

Why it matters. It lets warning exposure be compared between 2024 and 2025, showing whether a province's exposure is changing over time.

How it is used. Tracked separately. It does **not** replace the two-axis segmentation (above-median n_severe and above-median upside_swing) that sets the groups.

Bottom line: the project provides a transparent way to decide where closer observation and better data could be most useful.